

C LANGUAGE - ARRAYS EXERCISES

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Level 1: One-Dimensional Arrays (Basics)

1. Declare an array and display its elements.
2. Read n elements and print them in the same order.
3. Read n elements and print them in reverse order.
4. Find the sum of all elements in an array.
5. Find the average of all elements.
6. Find the maximum and minimum elements in an array.
7. Count even and odd numbers in an array.
8. Count positive and negative numbers in an array.
9. Search for a given element in an array (linear search).
10. Count the frequency of a given number.

Level 2: Intermediate Operations on Arrays

11. Copy one array into another.
12. Merge two arrays into a third array.
13. Sort an array in ascending order (Bubble Sort).
14. Sort an array in descending order.
15. Find the second largest element.
16. Remove duplicates from an array.
17. Insert an element at a specific position.
18. Delete an element from a specific position.
19. Rotate an array to the left by one position.
20. Rotate an array to the right by one position.

Level 3: Two-Dimensional Arrays (Matrices)

21. Read and display a matrix (2D array).
22. Find the sum of all elements in a matrix.
23. Find the sum of each row and each column.
24. Find the transpose of a matrix.
25. Add two matrices.
26. Subtract two matrices.
27. Multiply two matrices.
28. Check if a matrix is symmetric.
29. Find the sum of the main diagonal and secondary diagonal.
30. Find the upper and lower triangular parts of a square matrix.